W09

Spatial autocorrelation and point patterns

STATISTICS

Nearby points are more alike than distant ones (Tobler's first law). That is ecological information, but also a statistical problem: the data are not independent. Today you measure spatial autocorrelation and test whether nests are clustered, random or regularly spaced.

DURATION	About 2 hours.
PACKAGES	`spdep`, `gstat`, `spatstat`, `sf`, `tidyverse`
FILES	`week09_spatial_autocorrelation.R`, `week09_solutions.R`, `survey_points.csv`, `bird_counts_wide.csv`, `montfred.gpkg`
DATA	Abundance at the 40 points and 159 nests of two species in a 2 x 2 km beech forest plot.

Learning outcomes

- define neighbours and spatial weights;
- calculate and interpret Moran's I and a correlogram;
- fit a semivariogram and read nugget, sill and range;
- check whether a model's residuals are still autocorrelated;
- analyse point patterns with `spatstat`: intensity, quadrat test and Ripley's L function.

1 · Moran's I

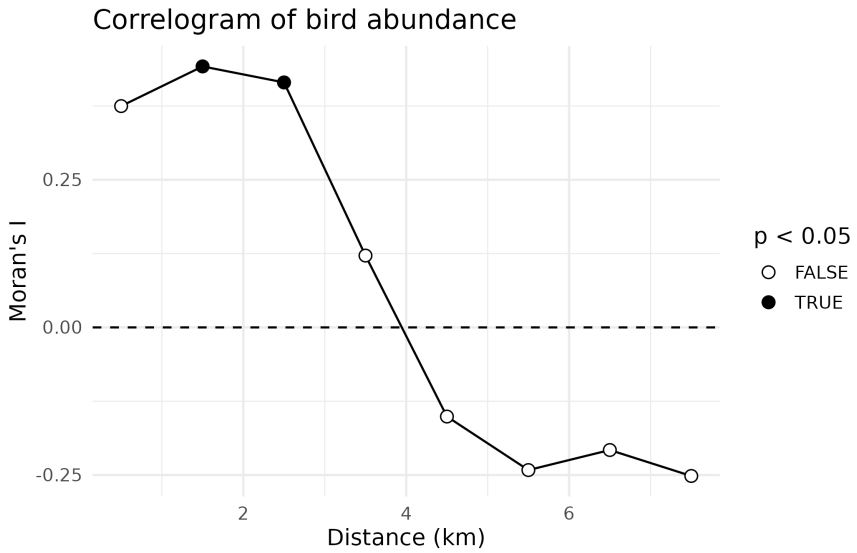
First decide who neighbours whom (here, points less than 2.5 km apart) and turn those links into weights. Moran's I compares each value with the mean of its neighbours: I > 0 means clustering (similar values together), I near 0 no pattern, and I < 0 alternation.

```
nb <- dnearneigh(xy, 0, 2500)                 # neighbours within 2.5 km
lw <- nb2listw(nb, style = "W", zero.policy = TRUE)
moran.test(pts$abundance, lw, zero.policy = TRUE)
moran.test(resid(lm(abundance ~ elevation_m, data = pts)), lw, zero.policy = TRUE)
```

Variable	Moran's I	p
Abundance	0.52	< 0.001
Residuals of the elevation model	0.13	0.10

Abundance is clearly clustered, but elevation explains almost all of that structure: in the residuals the index drops to 0.13 and is no longer significant. If residuals were still autocorrelated, you would need spatial

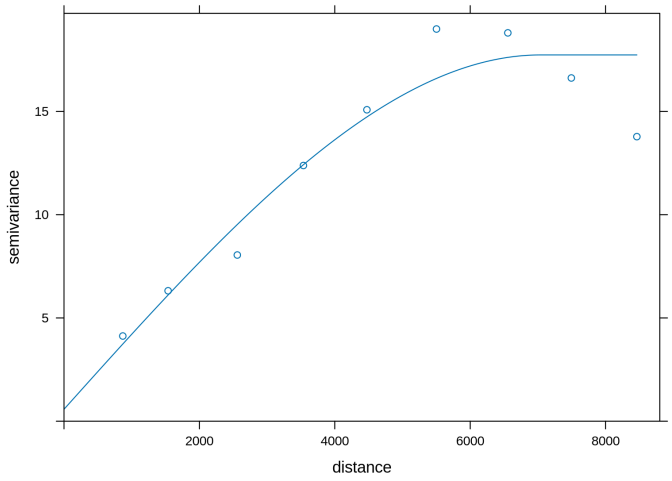
models (for example a spatial term in a GAM or a spatial mixed model).



Correlogram: Moran's I in 1 km bands. Points are similar up to about 3 km; beyond that the index turns negative (distant points differ).

2 · Semivariogram

The semivariogram measures how different values are as a function of distance. The **nugget** is the variation at near-zero distance (measurement error, micro-scale), the **sill** is the total variation and the **range** is the distance beyond which values are no more alike than expected. Here the spherical model gives a nugget of 0.6, a sill of about 18 and a range of about 7 km: the scale of the mountain's elevation gradient.



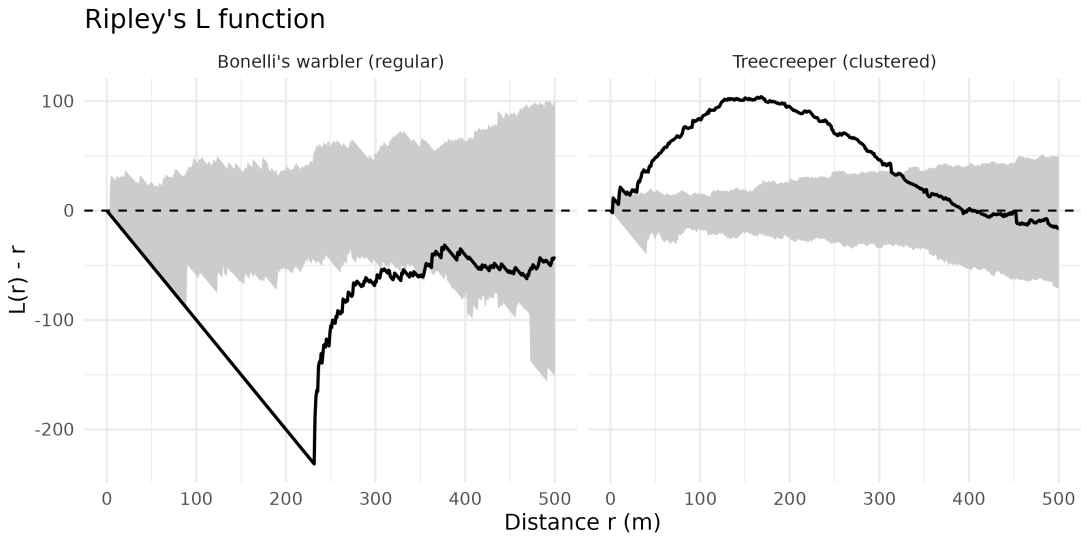
Empirical semivariogram of abundance (points) and fitted spherical model (line).

3 · Point patterns

Nests of two species were mapped in a 2 x 2 km plot. Short-toed treecreepers breed in groups tied to stands of old trees; Bonelli's warblers defend territories. Does it show in the pattern?

	Treecreeper	Bonelli's warbler
Nests	111	48
Intensity (nests/km²)	27.8	12.0
Quadrat test (4 x 4)	p < 0.001 (clustered)	p = 0.002 (too even)

Ripley's L function shows it at all scales at once. `L(r) - r` above the band of random simulations means clustering at that distance; below it, regularity. The treecreeper is clustered up to about 300 m; the warbler is regular below about 300 m, the spacing between its territories.



Ripley's L function with envelopes from 99 simulations of a random pattern (grey band).

Exercises

Answers in `week09_solutions.R`.

1. Repeat Moran's I for the residuals with the 4 nearest neighbours (`knearneigh(xy, k = 4)`). Does the conclusion change?
2. Moran's I for species richness.
3. Fit the variogram with an exponential model. What range do you get?
4. Mean nearest-nest distance (`nndist()`) for each species.
5. Use the G function (`Gest`) instead of L. Same conclusion?

Before Week 10

- Install `install.packages("maxnet")`.
- Have `gbif_lullula_arborea_clean.csv` from Week 8 ready and download `tavg.tif` and `prec.tif`.